Sahaj Bhadja

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Summary

Cloud Infrastructure Engineer with 2+ years building **Python automation**, **AWS/Azure systems**, containerized services, CI/CD pipelines, and reliable production platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Structures, Computer Networks

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built fault-tolerant FastAPI workers with explicit job states, timeouts, failure isolation, and monitoring, processing 1K+ jobs at 3x throughput while strengthening **production reliability and operational automation across services**.
- Architected multi-tenant Azure SQL and cloud-storage infrastructure for 500+ users, reducing dashboard latency 71% through **secure isolation, indexed queries, workload profiling, and scalable platform design across production services**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Engineered concurrent Python ingestion with ThreadPoolExecutor, OCR, timeouts, and model fallbacks, processing 1K+ files per batch while reducing turnaround 60% through **testing, profiling, and fault isolation under peak load**.
- Supported 10K+ daily requests through React and Flask services, resolving production defects and validating releases with **REST APIs, structured debugging, automated tests, and cross-functional deployment coordination across environments**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*
- Optimized distributed APIs and data pipelines, improving response times 30% and pipeline efficiency 68% through **AWS Glue, S3, caching, validation, and production performance analysis across recurring workloads**.
- Automated builds and 20+ unit and integration suites across Windows, Linux, and macOS using Jenkins, PyTest, Mockito, and virtual machines, strengthening **CI/CD, regression coverage, and release reliability**.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*
- Built eight microservices for authentication, resources, billing, permissions, and notifications, synchronizing 2M+ records through **REST APIs, WebSockets, scalable service boundaries, and reliable distributed production workflows across locations**.
- Automated Docker packaging and Jenkins releases, cutting deployment time 50% while using Sentry, Azure Monitor, and Kibana for **observability, incident investigation, and dependable production operations across services**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Built Flask and Firebase services for booking, pricing, payment, and status workflows across 50K+ records, maintaining sub-200-millisecond responses through **cloud integration, real-time synchronization, and performance validation under production load**.
- Mapped 12+ backend integrations and presented system architecture to client stakeholders before implementation, reducing design errors 25% through **technical documentation, collaborative review, and proactive early risk identification**.

Skills

Languages: Python, Bash, Go, TypeScript, JavaScript, Java, SQL, YAML, JSON
Cloud & IaC: AWS, Azure, GCP, Terraform, Terragrunt, Infrastructure as Code, IAM, Networking, Cost Optimization
Containers & Delivery: Kubernetes, Docker, ECS, Fargate, CI/CD, Jenkins, GitHub Actions, Git, Linux, Microservices
Operations: Prometheus, Grafana, Azure Monitor, Sentry, Kibana, Observability, Metrics, Logging, Alerting, SLOs, Incident Response, Runbooks
Engineering: FastAPI, Flask, PostgreSQL, Redis, Unit Testing, Integration Testing, Debugging, Root-Cause Analysis, Security, AI Coding Tools

Projects

InfraFabric | *Terraform, Terragrunt, Python, Bash, AWS, Azure, Kubernetes*
- Designed **reusable multi-cloud infrastructure abstractions** for networks, clusters, identities, and application environments with policy checks, cost tags, drift detection, auditable plans, and automated provisioning across AWS and Azure.

SignalGrid | *Python, Go, Prometheus, Grafana, Kubernetes, PagerDuty*
- Built a **production reliability and incident-response platform** with SLO dashboards, actionable alerts, failure injection, automated diagnostics, remediation workflows, and runbooks that convert recurring incidents into durable infrastructure improvements.

Linguistix | *Python, AWS ECS/Fargate, Docker, CodePipeline, CI/CD*
- Developed and deployed a **containerized data-processing API** on AWS ECS/Fargate with CodePipeline-driven CI/CD, automated validation checks, release controls, and production-grade reliability across environments while achieving 87% processing accuracy.